

Task 2: Data Analysis Using Pandas

Objective:

To demonstrate data analysis skills using Python Pandas library by loading, cleaning, filtering, grouping, and extracting insights from a dataset.

Dataset Description:

The dataset contains information about employees including:

- Name
- Age
- Department
- Salary

Some values were missing and were handled during data cleaning.

Tools Used:

- Python
- Pandas Library
- Spyder IDE

Steps Performed:

1. Importing Library

```
import pandas as pd
```

2. Loading Dataset

```
df = pd.read_csv("data.csv")  
print(df)
```

Output:

```
In [1]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task2_.py'  
--wdir  
   Name  Age  Department  Salary  
0  Amit  21.0         IT  30000.0  
1  Riya  22.0         HR  35000.0  
2   Raj  23.0         IT  40000.0  
3  Neha   NaN    Finance  45000.0  
4  Karan  25.0         IT  50000.0  
5 Simran  24.0         HR      NaN
```

3. Data Inspection

```
print(df.head())  
print(df.info())  
print(df.describe())
```

Output:

```
In [2]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task2_.py'  
--wdir  
   Name  Age  Department  Salary  
0  Amit  21.0         IT  30000.0  
1  Riya  22.0         HR  35000.0  
2   Raj  23.0         IT  40000.0  
3  Neha   NaN      Finance  45000.0  
4  Karan  25.0         IT  50000.0  
5 Simran  24.0         HR      NaN  
   Name  Age  Department  Salary  
0  Amit  21.0         IT  30000.0  
1  Riya  22.0         HR  35000.0  
2   Raj  23.0         IT  40000.0  
3  Neha   NaN      Finance  45000.0  
4  Karan  25.0         IT  50000.0  
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 6 entries, 0 to 5  
Data columns (total 4 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   Name        6 non-null      object  
1   Age          5 non-null      float64  
2   Department   6 non-null      object  
3   Salary       5 non-null      float64  
dtypes: float64(2), object(2)  
memory usage: 324.0+ bytes  
None  
Name          0  
Age           1  
Department     0  
Salary        1  
dtype: int64
```

4. Checking Missing Values

```
print(df.isnull().sum())
```

Ouput:

```
Name          0  
Age           0  
Department     0  
Salary        0  
dtype: int64
```

5. Data Cleaning

Missing values were handled using:

- Mean for numeric columns (Age, Salary)
- Mode for categorical columns (if required)

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
```

Output:

```
Empty DataFrame
Columns: [Name, Age, Department, Salary]
Index: []
```

6. Filtering Data

```
filtered_data = df[df["Age"] > 25]
```

```
print(filtered_data)
```

Output:

```
Department
Finance    45000.0
HR          37500.0
IT          40000.0
Name: Salary, dtype: float64
```

7. Grouping Data

```
grouped_data = df.groupby("Department")["Salary"].mean()
print(grouped_data)
```

Output:

```
In [8]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task2_.py'
--wdir
   Name  Age Department  Salary
0  Amit  21.0         IT  30000.0
1  Riya  22.0         HR  35000.0
2   Raj  23.0         IT  40000.0
3  Neha   NaN    Finance  45000.0
4  Karan  25.0         IT  50000.0
5 Simran  24.0         HR      NaN
Department
Finance    45000.0
HR         35000.0
IT         40000.0
Name: Salary, dtype: float64
```

8. Additional Insights

```
print(df["Salary"].max())
print(df["Department"].mode()[0])
```

Output:

```
In [7]: %runfile 'C:/Users/DELL USER/OneDrive/Desktop/HackathonProject/task2_.py'
--wdir
   Name  Age Department  Salary
0  Amit  21.0         IT  30000.0
1  Riya  22.0         HR  35000.0
2   Raj  23.0         IT  40000.0
3  Neha   NaN    Finance  45000.0
4  Karan  25.0         IT  50000.0
5 Simran  24.0         HR      NaN
Max Salary: 50000.0
Most common department: IT
```

Key Insights:

- Dataset contained missing values which were successfully handled.
- IT employees showed different salary trends compared to HR and Finance.
- Highest salary in dataset was identified.
- Most common department was found using mode function.
- Data cleaning improved accuracy of analysis.

Conclusion:

This analysis helped understand how to use Pandas for real-world data processing. It demonstrated data loading, cleaning, filtering, grouping, and extracting meaningful insights efficiently.